Parseval's Identity for Fourier Series

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

A complete authored walkthrough of parseval's identity for fourier series, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

HA-03 / Harmonic Analysis / Report v1

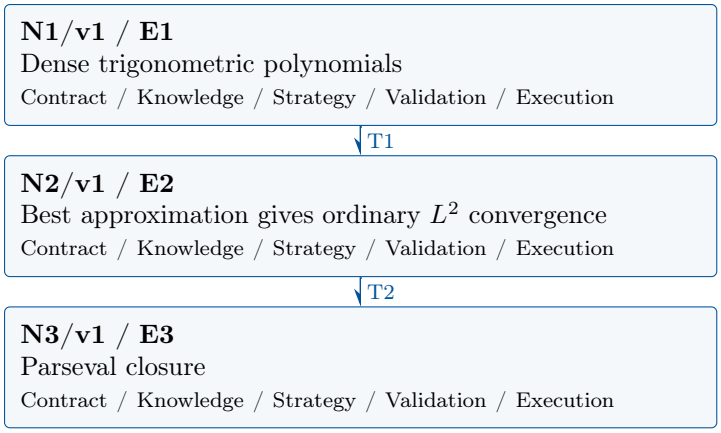
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Under HA-01's normalization, prove for every $f \in L^2(\mathbb{T})$ that Fourier partial sums converge in L^2 and $\|f\|_2^2 = \sum_{n\in\mathbb{Z}} |c_n|^2$.*

A. Foundations and permitted imports

Import HA-01's finite projection identity and HA-02: Cesàro means of Fourier sums approximate every continuous periodic function uniformly. Also import density of continuous functions in $L^2(T)$ for normalized Lebesgue measure (obtainable from simple functions and approximation of measurable sets). These imports imply trigonometric-polynomial density, not pointwise convergence of ordinary sums.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. Every L^2 function can be approximated in L^2 by a trigonometric polynomial.

Receiving activity. N2: Best approximation gives ordinary L^2 convergence

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Density and HA-01 projection identity; prior exports E1/v1

T2: N2/E2/v1 to N3. Ordinary Fourier partial sums converge to f in L^2.

Receiving activity. N3: Parseval closure

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Vanishing projection residual; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Dense trigonometric polynomials

Object and role. Combine two approximation errors.

Inputs. Continuous density in L^2; uniform Fejér approximation; foundational imports stated in §1

C — Contract

Own E1/v1: Every L^2 function can be approximated in L^2 by a trigonometric polynomial.

K — Knowledge

Retain Continuous density in L^2; uniform Fejér approximation; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Combine two approximation errors.

V — Validation

Uniform Fejér convergence is used only for continuous h, not arbitrary L^2 f.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 For $f \in L^2$ and $\varepsilon > 0$ choose continuous h with $\|f - h\|_2 < \varepsilon/2$.

N1.2 Choose a Fejér trigonometric polynomial p with $\|h - p\|\infty < \varepsilon/2$.

N1.3 Since measure$(T) = 1$, $\|h - p\|_2 \leq \|h - p\|\infty$.

N1.4 Thus $\|f - p\|_2 < \varepsilon$.

N1.5 The polynomial space is dense in L^2.

Export E1/v1. Every L^2 function can be approximated in L^2 by a trigonometric polynomial.

Acceptance gate. Uniform Fejér convergence is used only for continuous h, not arbitrary L^2 f. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Uniform Fejér convergence is used only for continuous h, not arbitrary L^2 f. Recheck dependent nodes N2, N3.

B. N2/v1: Best approximation gives ordinary L^2 convergence

Object and role. Use a fixed approximating polynomial inside all sufficiently large frequency spaces.

Inputs. Density and HA-01 projection identity; prior exports E1/v1

C — Contract

Own E2/v1: Ordinary Fourier partial sums converge to f in L^2.

K — Knowledge

Retain Density and HA-01 projection identity; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Use a fixed approximating polynomial inside all sufficiently large frequency spaces.

V — Validation

The conclusion is L^2, not pointwise or uniform convergence.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Let $S_N f = \sum_{|n|\le N} c_n e_n$.

N2.2 If p has frequencies at mostN, then $f - S_N f$ is orthogonal to $S_N f - p$.

N2.3 Hence $\|f-p\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - p\|_2^2$, so $\|f - S_N f\|_2 \le \|f-p\|_2$.

N2.4 Choose p within ε by the preceding node, then retain that p for every N at least its degree.

N2.5 Thus $S_N f \to f$ in L^2.

Export E2/v1. Ordinary Fourier partial sums converge to f in L^2.

Acceptance gate. The conclusion is L^2, not pointwise or uniform convergence. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: The conclusion is L^2, not pointwise or uniform convergence. Recheck dependent nodes N3.

C. N3/v1: Parseval closure

Object and role. Pass the exact finite norm identity to the limit.

Inputs. Vanishing projection residual; prior exports E2/v1

C — Contract

Own E3/v1: Parseval identity and L^2 Fourier reconstruction.

K — Knowledge

Retain Vanishing projection residual; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Pass the exact finite norm identity to the limit.

V — Validation

Normalization must remain consistent throughout both approximation steps.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 HA-01 gives $\|f\|_2^2 = \sum_{|n|\le N} |c_n|^2 + \|f - S_N f\|_2^2$.

N3.2 The residual tends to 0; the nonnegative coefficient sums converge.

N3.3 Therefore $\|f\|_2^2 = \sum_Z |c_n|^2$, completing both target statements.

Export E3/v1. Parseval identity and L^2 Fourier reconstruction.

Acceptance gate. Normalization must remain consistent throughout both approximation steps. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Normalization must remain consistent throughout both approximation steps. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Dense trigonometric polynomials

1. For $f \in L^2$ and $\varepsilon > 0$ choose continuous h with $\|f - h\|_2 < \varepsilon/2$.
2. Choose a Fejér trigonometric polynomial p with $\|h - p\|\infty < \varepsilon/2$.
3. Since measure$(T) = 1$, $\|h - p\|_2 \leq \|h - p\|\infty$.
4. Thus $\|f - p\|_2 < \varepsilon$.
5. The polynomial space is dense in L^2.

B. §4.N2 — Best approximation gives ordinary L^2 convergence

1. Let $S_N f = \sum_{|n| \leq N} c_n e_n$.
2. If p has frequencies at mostN, then $f - S_N f$ is orthogonal to $S_N f - p$.
3. Hence $\|f - p\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - p\|_2^2$, so $\|f - S_N f\|_2 \leq \|f - p\|_2$.
4. Choose p within ε by the preceding node, then retain that p for every N at least its degree.
5. Thus $S_N f \to f$ in L^2.

C. §4.N3 — Parseval closure

1. HA-01 gives $\|f\|_2^2 = \sum_{|n| \leq N} |c_n|^2 + \|f - S_N f\|_2^2$.
2. The residual tends to 0; the nonnegative coefficient sums converge.
3. Therefore $\|f\|_2^2 = \sum_Z |c_n|^2$, completing both target statements.

□

V. WORKED ORCHESTRATION

For the periodic square wave $f = 1$ on $(0, \pi)$ and -1 on $(-\pi, 0)$, $\|f\|_2^2 = 1$. Integration gives $c_n = 2/(\pi$ i n) for odd n and 0 for even n, including $c_0 = 0$. Parseval yields $1 = (8/\pi^2) \sum_{j \geq 0} 1/(2j+1)^2$, so the odd reciprocal-square sum is $\pi^2/8$.

A. Boundary and hypothesis audit

The square wave has jumps; uniform convergence of its continuous Fourier partial sums is impossible. Validation accepts the L^2 export and rejects promoting it to uniform convergence. Fejér's theorem and ordinary partial-sum convergence concern different statements.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

The square wave has jumps; uniform convergence of its continuous Fourier partial sums is impossible. Validation accepts the L^2 export and rejects promoting it to uniform convergence. Fejér's theorem and ordinary partial-sum convergence concern different statements.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.